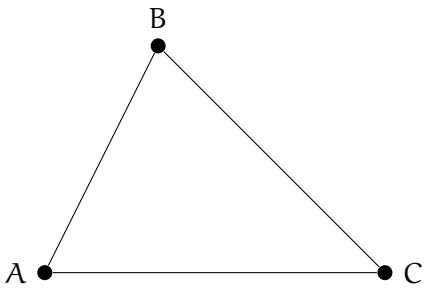
CHAPTER 1

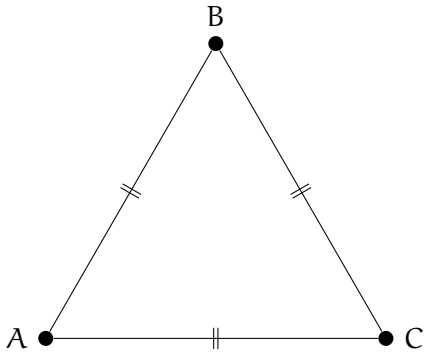
# Introduction to Triangles

Connecting any three points with three line segments will get you a triangle. Here is the triangle $ABC$, which was created by connecting three points $A$, $B$, and $C$:

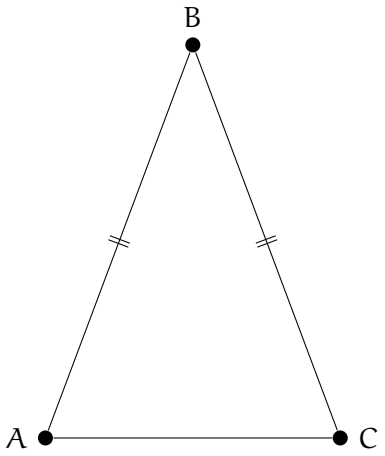



## 1.1 Equilateral, Isosceles, and Scalene Triangles

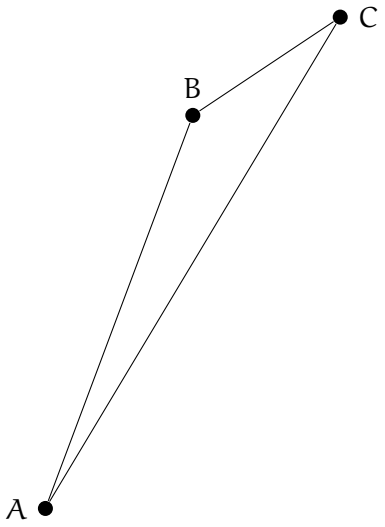
We talk a great deal about the length of the sides of triangles. If all three sides of the triangle are the same length, we say it is an *equilateral triangle*:




If only two sides of the triangle are the same length, we say it is an *isosceles triangle*:

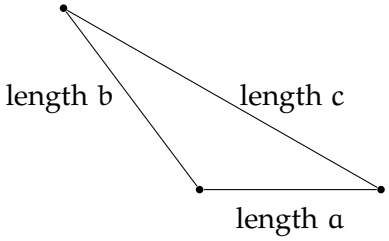



When all three sides of a triangle are different, the triangle is referred to as *scalene*. This consequently means all angle measures are different as well.



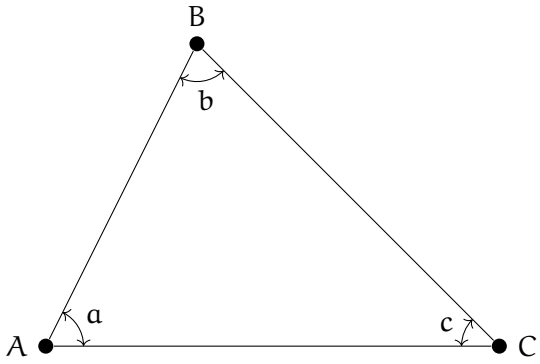

The shortest distance between two points is always the straight line between them. This means you can be certain that the length of one side will *always* be less than the sum of the lengths of the remaining two sides. This is known as the *triangle inequality*.

For example, in this diagram, $c$ must be less than $a + b$.

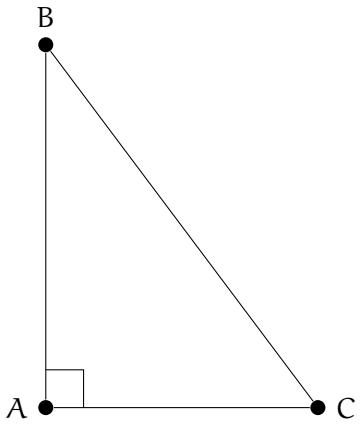



## 1.2 Interior Angles of a Triangle

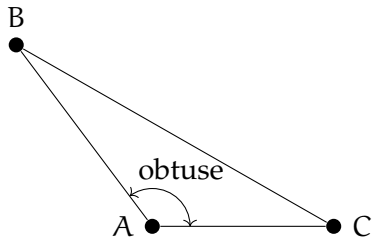
We also talk a lot about the interior angles of a triangle:




A triangle where one of the interior angles is a right angle is said to be a *right triangle*: The side opposite the right angle is the longest side, referred to as the hypotenuse.




If a triangle has an obtuse interior angle, it is said to be an *obtuse triangle*:




If all three interior angles of a triangle are less than 90°, it is said to be an *acute triangle*.

The measures of the interior angles of a triangle always add up to 180°. For example, if we know that a triangle has interior angles of 37° and 56°, we know that the third interior angle is 87°.

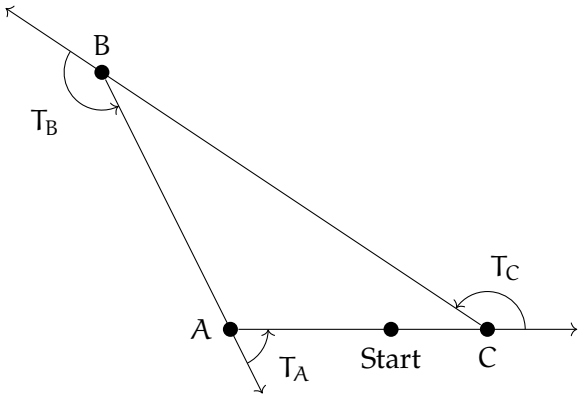
## *Exercise 1* Missing Angle

One interior angle of a triangle is 92°. The second angle is 42°. What is the measure of the third interior angle?

*Working Space*

*Answer on Page 7*

How can you know that the sum of the interior angles is 180°? Imagine that you started on the edge of a triangle and walked all the way around to where you started. ( going counter-clockwise.) You would turn three times to the left:




After these three turns, you would be facing the same direction that you started in. Thus, $T_A + T_B + T_C = 360°$. The measures of the interior angles are a, b, and c. Notice that a and $T_A$ are supplementary. So we know that:

- $T_A = 180 - a$
- $T_B = 180 - b$
- $T_C = 180 - c$

So we can rewrite the equation above as

$$(180 - a) + (180 - b) + (180 - c) = 360^\circ$$

Which simplifies to:

$$540^\circ - (a + b + c) = 360^\circ$$

Subtracting both sides from 540:

$$a + b + c = 180^\circ$$

## *Exercise 2* Interior Angles of a Quadrilateral

Any four-sided polygon is a *quadrilateral*. Using the same "walk around the edge" logic, what is the sum of the interior angles of any quadrilateral?

*Working Space*

*Answer on Page 7*

## 1.3 Is it an Equilateral, Isosceles, or Scalene Triangle with Python?

You have now seen three ways to classify a triangle by its sides, and one rule that decides whether three lengths make a triangle at all. A short Python function can apply both ideas to any triple of numbers you hand it.

```
1  def classify_triangle(a, b, c):
2      short1, short2, longest = sorted([a, b, c])
3      if short1 + short2 <= longest:
4          return "not a triangle"
5      if a == b == c:
6          return "equilateral"
7      if a == b or b == c or a == c:
8          return "isosceles"
9      return "scalene"
```

Sorting the three lengths puts the two shortest sides first, no matter what order they were given in. The triangle inequality check then compares those two shortest sides against the longest one, exactly as stated earlier in this chapter: the length of one side must always be less than the sum of the other two. Once that check passes, the function looks for three equal sides, then two, and falls back to scalene if neither pattern matches.

```
triangles = [(5, 5, 5), (5, 5, 8), (5, 12, 13), (2, 2, 5)]
for sides in triangles:
    print(sides, classify_triangle(*sides))
```

This prints:

```
(5, 5, 5) equilateral
(5, 5, 8) isosceles
(5, 12, 13) scalene
(2, 2, 5) not a triangle
```

Notice the last line. $2 + 2 = 4$, which is not greater than 5, so these three lengths cannot close into a triangle at all, and the function catches that before it ever checks for equal sides.

## *Exercise 3* Classifying Triangles

Classify each of these triples by hand, then check your answers by running them through `classify_triangle`: $(7, 7, 10)$, $(9, 40, 41)$, and $(3, 4, 9)$.

*Working Space*

*Answer on Page 7*

Appendix A

# Answers to Exercises

## Answer to Exercise 1 (on page 4)

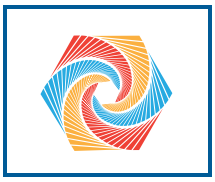
$180° - (92° + 42°) = 46°$

## Answer to Exercise 2 (on page 5)

$360°$

## Answer to Exercise 3 (on page 6)

$(7, 7, 10)$: two sides match, so this is isosceles. $(9, 40, 41)$: $9 + 40 = 49 > 41$ and no two sides match, so this is scalene. $(3, 4, 9)$: $3 + 4 = 7$, which is less than 9, so this is not a triangle.



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